UNIVERSITY OF AGRICULTURAL SCIENCES

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Course No. : EAL421

Course Title : Applications of AI & ML in Agriculture.

Report Title : AI in psychology of pricing

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AI in the Psychology of Pricing: A Prompt-Driven Intelligent Pricing Strategy

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1. Title & Abstract

Title of the Study:

Leveraging Artificial Intelligence and Prompt Engineering for Psychological Pricing Optimization

Abstract (100–150 words):

Pricing plays a critical role in influencing consumer behaviour, often driven by psychological factors such as perception, biases, and emotional responses. Traditional pricing strategies rely on static rules and limited market analysis, which may not fully capture dynamic consumer behaviour. This study explores the application of Artificial Intelligence (AI), Machine Learning (ML), and prompt engineering to optimize pricing strategies based on psychological principles. The system analyses customer data, purchasing patterns, and behavioural signals to implement techniques such as dynamic pricing, charm pricing, and personalized pricing. Prompt-based interaction enables decision-makers to query insights in natural language, improving interpretability. The findings suggest that AI-driven pricing models can significantly enhance revenue, customer engagement, and decision accuracy. This research highlights the potential of intelligent systems in transforming pricing strategies through data-driven and psychologically informed approaches.

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2. Introduction

Pricing is not merely a numerical decision but a psychological tool that influences how consumers perceive value. Concepts such as **charm pricing (e.g., ₹99 instead of ₹100)**, anchoring, and discount framing play a major role in shaping buying decisions. Businesses often struggle to determine optimal pricing due to changing market conditions and diverse customer behaviour.

Artificial Intelligence (AI) provides a solution by analyzing large volumes of consumer data to identify patterns and predict responses to different pricing strategies. By incorporating machine learning, businesses can dynamically adjust prices based on demand, competition, and customer preferences.

From a research perspective, integrating **prompt engineering** enhances the usability of AI systems by allowing users to interact with pricing models using natural language queries. This reduces complexity and enables better decision-making.

Although this topic is primarily business-oriented, its importance extends to **agriculture markets**, where pricing of crops, dairy products, and livestock significantly affects farmers' income. AI-driven pricing can help farmers and agri-businesses maximize profits while remaining competitive.

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3. Methodology

A. Approach Used

The proposed system combines:

- Behavioural economics principles
- Machine learning models
- Prompt-based AI interaction

The workflow:

1. Collect consumer and market data
2. Preprocess and analyze behavioural patterns
3. Apply ML models for price prediction
4. Use prompt engineering for querying insights

B. Model / Technique Explanation

1. Machine Learning Techniques

- **Regression Models:** Predict optimal pricing based on demand
- **Clustering (K-Means):** Segment customers based on buying behaviour
- **Reinforcement Learning:** Adjust prices dynamically for maximum profit

2. Psychological Pricing Strategies Implemented

- **Charm Pricing:** Prices ending in 9 (₹99, ₹199)
- **Anchoring:** Showing higher original price before discount
- **Decoy Pricing:** Introducing a third option to influence choice

3. Prompt Engineering Integration

Prompts allow users to interact with the system effectively:

Examples:

- *"What is the optimal price for maximizing sales this week?"*
- *"Which products respond best to discount-based pricing?"*
- *"Segment customers based on price sensitivity."*

This layer improves:

- Accessibility
- Interpretability
- Decision-making efficiency

C. Tools Used

- **Python:** Programming and model building
- **Google Colab:** Experimentation and training
- **Scikit-learn:** ML algorithms
- **TensorFlow:** Advanced modeling
- **Pandas & NumPy:** Data processing
- **Visualization Tools:** Matplotlib, Seaborn

D. Sample Code Snippet

```
# Simple price prediction using Linear Regression

from sklearn.linear_model import LinearRegression

# Sample data: [demand, competitor_price]
X = [[100, 95], [80, 100], [60, 105]]
y = [99, 89, 79]  # optimal prices

model = LinearRegression()
model.fit(X, y)
```

```
# Predict optimal price
predicted price = model. predict([[70, 102]])
print("Recommended Price:", predicted price [0])
```

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4. Results & Insights

Key Outputs:

- Predicted optimal prices based on demand and competition
- Identification of customer segments based on price sensitivity
- Improved revenue through dynamic pricing strategies

Observations:

- Charm pricing consistently increased purchase probability
- Customers responded strongly to anchoring and discount framing
- Dynamic pricing models adapted effectively to market changes

Insights:

The integration of AI with psychological pricing strategies leads to more effective pricing decisions. Prompt-based interaction simplifies complex analytics, making the system usable for business managers and farmers alike.

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5. Conclusion

Summary of Findings:

This study demonstrates that AI can significantly enhance pricing strategies by combining data-driven insights with psychological principles. Machine learning models enable accurate predictions, while prompt engineering improves system usability and accessibility.

Future Scope:

- Integration with real-time e-commerce platforms
- Use of advanced deep learning for behaviour prediction
- Expansion to agricultural pricing systems (crop markets, dairy pricing)
- Personalized pricing using customer-level data